Quiz — 1.3.3 Networks — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks, 1 each)

1. **B** — LAN = small area, single site, organisation-owned hardware.
2. **B** — In P2P all peers are equal, acting as both client and server.
3. **B** — The central switch/hub is the single point of failure in a star.
4. **B** — Packets are routed independently and reassembled at the destination.
5. **C** — SMTP sends mail; POP3/IMAP retrieve it.
6. **B** — DNS translates domain names to IP addresses.
7. **B** — A router connects different networks and routes using IP.
8. **C** — The Network/Internet layer adds IP addressing and routes packets.

Section B (12 marks)

9. **[2]** Any two for 1 mark each: centralised security/backups; easier to manage/update centrally; scales well to many users; central data storage/control.

10. **[4] (1 mark each)** - A → **Link / Data link** layer - B → **Network / Internet** layer - C → **Transport** layer - D → **Application** layer

11. **[2]** 1 mark: a switch connects devices *within* a single LAN and forwards frames using MAC addresses. 1 mark: a router connects *different* networks (e.g. LAN to internet) and forwards packets using IP addresses.

12. **[2]** 1 mark: a proxy acts as an intermediary between client and server, forwarding requests/responses. 1 mark for one benefit: hides the user's IP address, caches pages to speed up access, or filters/logs/monitors traffic.

13. **[2]** 1 mark: packets can be routed independently along different routes. 1 mark: if one route/node fails, packets can be re-routed around it (no single dedicated path to break), so the communication still gets through.

Section C (5 marks — award up to 5 from)

14.
    - DNS translates the typed domain name into the server's IP address (querying DNS servers, results may be cached). **[1]**
    - The browser (client) sends a request to that IP address; HTTPS is used so the data is encrypted in transit. **[1]**
    - The data is split into packets at the transport layer, each given a header (e.g. sequence number, ports). **[1]**
    - Packets are given source/destination IP addresses and routed independently across the network, then reassembled in order at the destination. **[1]**
    - A router (and/or switch/NIC) forwards the packets between networks toward the destination using IP addresses. **[1]**

Total: 8 + 12 + 5 = 25 marks